

Bounded Archive Expansion in the Structura Reditus Intake Pipeline: Reliability Under Nested Depth

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This manuscript is a synthetic acceptance fixture: it describes only the research library's own intake machinery.

Abstract

We test whether the intake pipeline's bounded ZIP expansion returns the same member set under repeated runs. In a simulation of two hundred synthetic archives the simulated pipeline returned identical member sets in every run; one archive with a nested depth of four failed as designed. The failure is reported as a result, not as missing information. This revision adds the nested-depth analysis.

Introduction

The pipeline is a diagnostic of the library's own robustness. Bounded expansion, member-count limits and a time budget are the contract under test; nothing here is a scholarly claim about any external domain.

Method

Two hundred synthetic archives were generated with controlled depth and member counts and passed through the expansion routine ten times each. Identity of the returned member set was checked by hashing.

Result

All archives within the contract returned identical member sets on every run. The single archive nested beyond depth three was refused each time, as the contract requires.